

Smart PO-to-Budget Linking System

CareWare Procurement Integration with Budget Management System

1. Project Objective

Business Problem

Today, when a Purchase Order (PO) is created in CareWare, the Purchasing Department must manually decide which approved budget item should pay for that PO.

Example:

PO #2208

Item:

Stand Alone AC

Amount:

13,660 SAR

The purchasing employee must manually search:

Department Budget

↓

Budget Category

↓

Budget Item

Then manually link the PO.

Problems Today

Time Consuming

Every PO requires manual searching.

Human Error

Users may choose the wrong budget item.

Inconsistent Decisions

Different users may classify similar purchases differently.

Example:

```
User A:  
Stand Alone AC  
→ Air Conditioning  
  
User B:  
Stand Alone AC  
→ Building Maintenance
```

No Learning

The system never remembers previous decisions.

The same work is repeated again and again.

2. Proposed Solution

The Budget System will become an intelligent assistant.

Instead of forcing the user to search manually, the system will suggest the most likely approved budget item.

The user remains in full control.

The system only provides recommendations.

3. Complete Future Workflow

```
PO Created in CareWare  
↓  
Automatic Sync  
↓  
PO Waiting For Budget Link  
↓  
Budget Suggestion Engine  
↓  
Suggested Budget Item  
↓  
User Review  
↓  
Approve / Change
```

↓
Budget Commitment Created
↓
System Learns Decision

4. Phase 1 – CareWare Synchronization

Business Goal

Automatically retrieve newly approved POs from CareWare.

No Excel.

No manual import.

Technical Process

Every 30 minutes:

Budget System
↓
Query Oracle
↓
Get New POs
↓
Store Locally

Example:

PO #2208

Item:
Stand Alone AC

Amount:
13,660 SAR

Result

The PO appears inside the Budget System.

Status:

Waiting For Budget Link

No budget impact yet.

5. Phase 2 – Budget Suggestion Engine

Business Goal

Answer one question:

Which approved budget item should pay for this PO?

Example Budget

Engineering Budget

Maintenance

- └ Air Conditioning
- └ Electrical Works
- └ Plumbing

Furniture

- └ Office Chairs
- └ Office Desks

New PO:

Stand Alone AC

The system must determine:

Air Conditioning?
Electrical Works?
Office Chairs?

6. Phase 3 – Exact Historical Matching

Business Goal

Reuse previous approvals.

What Data Is Used?

Table:

PO Approval History

Example:

PO Item	Budget Item
Stand Alone AC	Air Conditioning

Step-by-Step Process

Step 1

New PO arrives:

Stand Alone AC

Step 2

System searches approval history.

Question:

Has this exact item been approved before?

Step 3

Result:

YES

Found:

Stand Alone AC
↓
Air Conditioning

Step 4

System creates suggestion:

Maintenance
↓
Air Conditioning

Step 5

User sees:

Suggested Budget Item:
Maintenance → Air Conditioning

Reason:
Previously approved

Step 6

User clicks:

Approve

Why Confidence Is 100%?

Because a human already approved this exact item previously.

No guessing is involved.

7. Phase 4 – Similar Historical PO Matching

Business Goal

Handle items that have never been approved before.

Example

New PO:

Portable Cooling Unit

Step 1

System searches:

Has Portable Cooling Unit
been approved before?

Result:

NO

Step 2

System searches previously approved PO items.

Example History:

Previous PO Item	Budget Item
Stand Alone AC	Air Conditioning
Window AC	Air Conditioning
Split AC	Air Conditioning
Central AC	Air Conditioning
Office Chair	Office Chairs
Network Switch	IT Equipment

Step 3

System compares the new PO description against previous approved PO descriptions.

Example:

Portable Cooling Unit

compared with:

Stand Alone AC
Window AC
Split AC
Central AC

The comparison is based on semantic similarity, meaning the system compares the meaning of the descriptions rather than exact keywords. Embedding models represent text as numerical vectors and similar meanings are located closer together in vector space.

Step 4

System finds the most similar historical approvals.

Example:

Similar Historical Item	Linked Budget Item
Split AC	Air Conditioning
Window AC	Air Conditioning
Stand Alone AC	Air Conditioning

Step 5

System asks:

What budget item
were these similar purchases linked to?

Answer:

Air Conditioning
Air Conditioning
Air Conditioning

Step 6

System creates suggestion:

```
Maintenance
↓
Air Conditioning
```

What User Sees?

```
Suggested Budget Item:
Maintenance → Air Conditioning

Reason:
3 similar purchases were previously approved under Air Conditioning
```

This explanation is far more useful than showing:

```
Confidence = 95%
```

alone.

8. Phase 5 – AI Assistance

Business Goal

Handle completely new items.

Example:

```
Portable Vaccine Storage Refrigerator
```

No exact match.

No similar approved purchases.

The system sends:

```
PO Description
Supplier
```

Department
Available Budget Items

to the AI service.

AI returns:

Suggested:
Medical Equipment

Reason:
Medical refrigeration equipment is typically categorized as medical equipment.

User still makes the final decision.

AI never approves automatically.

9. Phase 6 – User Review Screen

User sees:

PO	Suggested Budget Item	Reason
Stand Alone AC	Air Conditioning	Previously approved
Portable Cooling Unit	Air Conditioning	Similar approved purchases
Vaccine Refrigerator	Medical Equipment	AI recommendation

Actions:

Approve
Change
Reject

10. Phase 7 – Budget Commitment Creation

After approval:

PO #2208

13,660 SAR

linked to:

Maintenance
↓
Air Conditioning

Budget Before:

Type	Amount
Approved	100,000
Committed	0
Available	100,000

Budget After:

Type	Amount
Approved	100,000
Committed	13,660
Available	86,340

11. Phase 8 – Learning System

Every approval becomes future knowledge.

Example:

Portable Cooling Unit
↓
Air Conditioning

saved permanently.

Next time:

Portable Cooling Unit

arrives.

The system skips Phase 4 and immediately suggests:

Air Conditioning

because it already learned from the previous approval.

12. Phase 9 – Actual Spending Tracking

When CareWare reports:

Invoice Posted

or

Goods Received

the system converts:

Committed Amount

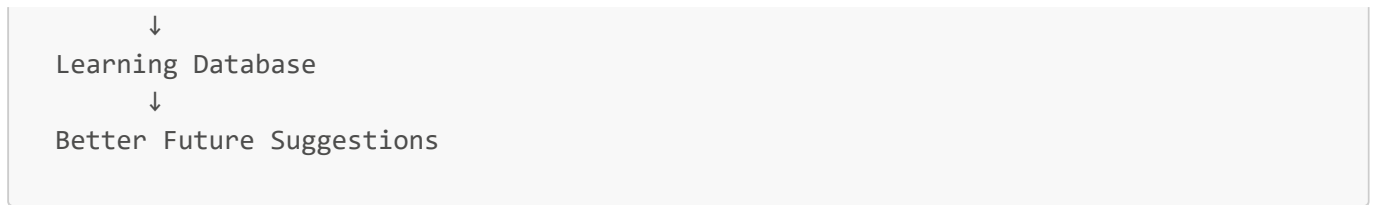
into:

Actual Spending

keeping budget balances accurate.

Final Vision

```
graph TD
    A[CareWare PO] --> B[Automatic Budget Item Suggestion]
    B --> C[User Approval]
    C --> D[Budget Commitment]
```



The system does not replace purchasing staff.

The system acts as an intelligent procurement assistant that reduces manual work, improves consistency, and continuously learns from real purchasing decisions.